

Breast Ultrasound Image Classification using Deep Learning

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1 Introduction

Breast cancer is one of the most common and life-threatening diseases affecting women worldwide. Early detection significantly increases the chances of successful treatment and survival. Medical imaging techniques such as mammography, MRI, and ultrasound are widely used for breast cancer diagnosis.

Among these methods, **ultrasound imaging** is commonly used because it is safe, cost-effective, and does not involve radiation exposure. However, manual interpretation of ultrasound images can be challenging and requires expert radiologists.

Recent advances in **deep learning and computer vision** have enabled automated analysis of medical images. Convolutional Neural Networks (CNNs) can learn complex visual patterns from medical data and assist doctors in making accurate diagnoses.

The objective of this project is to develop a **deep learning based classification system** that can automatically classify breast ultrasound images into three categories:

- **Normal**
- **Benign**
- **Malignant**

The system is trained and evaluated using the **BUSI (Breast Ultrasound Images) dataset**. Different techniques are explored to handle **class imbalance**, including:

- Class weighting
- Oversampling

- Data augmentation
- Focal loss

The performance of each method is compared using standard evaluation metrics.

2 Dataset Description

The dataset used in this project is the **BUSI (Breast Ultrasound Images) Dataset**.

The dataset contains ultrasound images categorized into three different medical conditions.

2.1 Classes in Dataset

Class	Description
Normal	Healthy breast tissue
Benign	Non-cancerous tumor
Malignant	Cancerous tumor

2.2 Dataset Characteristics

- Total images: ~ 780
- Number of classes: 3
- Image type: Ultrasound images
- Format: PNG images
- Dataset used through PyTorch `ImageFolder`

The dataset shows **class imbalance**, meaning that some classes contain more images than others. This imbalance can negatively affect model training because the model may become biased toward the majority class.

To address this issue, several techniques were implemented during training.

3 Data Preprocessing

Before training the model, several preprocessing steps were applied to the dataset.

3.1 Image Processing Steps

The following transformations were applied to prepare the images:

- Image resizing to **224 × 224 pixels**
- Conversion of images to **PyTorch tensors**
- Pixel value **normalization**

These steps ensure that all images have a consistent format suitable for training deep learning models.

4 Dataset Splitting

The dataset was divided into three subsets:

Dataset	Purpose
Training Set	Used to train the neural network
Validation Set	Used for tuning the model
Test Set	Used for final evaluation

A typical split was used:

- **70% Training**
- **15% Validation**
- **15% Testing**

The test dataset contained **237 images** used to evaluate final performance.

5 Model Architecture

In this project, **ResNet-18** was used as the base deep learning model.

ResNet (Residual Network) is a convolutional neural network architecture that introduces **residual connections**. These connections allow gradients to flow easily during training and help prevent the vanishing gradient problem.

5.1 Key Properties of ResNet-18

- 18 deep layers
- Residual skip connections
- Pretrained on ImageNet
- Effective for image classification tasks

The final fully connected layer of ResNet-18 was modified to output **three classes** corresponding to:

- Normal
- Benign
- Malignant

6 Baseline Model

A baseline model was first trained using the original dataset without applying any imbalance handling techniques.

6.1 Baseline Model Results

Class	Precision	Recall	F1 Score
Normal	0.98	0.89	0.93
Benign	0.88	0.97	0.92
Malignant	0.85	1.00	0.92

Overall Accuracy: 0.93

Macro F1 Score: 0.9257

Weighted F1 Score: 0.9286

The baseline model achieved good performance but could still be affected by class imbalance.

7 Handling Class Imbalance

Medical image datasets frequently contain imbalanced classes. To improve model performance, several techniques were tested.

7.1 Class Weighting

Class weighting modifies the loss function by assigning higher importance to minority classes.

7.2 Results with Class Weighting

Class	Precision	Recall	F1 Score
Normal	0.97	0.90	0.93
Benign	0.91	0.95	0.93
Malignant	0.83	0.95	0.88

Accuracy: 0.92

Macro F1: 0.9161

Weighted F1: 0.9247

Class weighting slightly balanced performance across classes.

7.3 Oversampling

Oversampling increases the representation of minority classes during training. A **Weighted Random Sampler** was used so that underrepresented classes appear more frequently in training batches.

7.4 Results with Oversampling

Class	Precision	Recall	F1 Score
Normal	0.95	0.91	0.93
Benign	0.96	0.87	0.92
Malignant	0.78	1.00	0.88

Accuracy: 0.92

Macro F1: 0.9078

Weighted F1: 0.9166

Oversampling helped improve recall for the minority class but slightly reduced precision.

7.5 Data Augmentation

Data augmentation artificially increases the diversity of the training dataset by applying transformations to images.

Examples of augmentations used include:

- Random horizontal flip
- Random rotation
- Random cropping
- Image normalization

7.6 Results with Data Augmentation

Class	Precision	Recall	F1 Score
Normal	0.94	0.97	0.96
Benign	1.00	0.90	0.95
Malignant	0.90	0.95	0.93

Accuracy: 0.95

Macro F1: 0.9442

Weighted F1: 0.9494

Data augmentation produced the **best overall performance** among all tested methods.

8 Evaluation Metrics

To measure model performance, several evaluation metrics are used.

8.1 Accuracy

Accuracy measures the overall percentage of correctly classified images.

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

8.2 Precision

Precision measures how many predicted positives are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

8.3 Recall

Recall measures how many actual positives were correctly detected.

$$Recall = \frac{TP}{TP + FN}$$

8.4 F1 Score

The F1 score is the harmonic mean of precision and recall.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

8.5 Classification Report

A classification report provides detailed metrics for each class including:

- Precision
- Recall
- F1 Score
- Support

These metrics help evaluate model performance for each category individually.

9 Results Comparison

Method	Macro F1	Weighted F1
Baseline	0.9257	0.9286
Class Weighting	0.9161	0.9247
Oversampling	0.9078	0.9166
Data Augmentation	0.9442	0.9494
Focal Loss	0.8552	0.8487

From the comparison, **data augmentation achieved the best overall performance.**

10 Conclusion

This project developed a deep learning system for classifying breast ultrasound images using the BUSI dataset.

A **ResNet-18 convolutional neural network** was trained to classify images into three categories: Normal, Benign, and Malignant.

Several strategies were tested to address class imbalance, including:

- Class weighting
- Oversampling
- Data augmentation
- Focal loss

Among these approaches, **data augmentation produced the best results**, achieving the highest macro F1 and weighted F1 scores.

The results demonstrate that deep learning techniques can effectively classify medical images and may assist radiologists in early breast cancer detection.

Future improvements may include:

- Using deeper architectures (ResNet50, EfficientNet)
- Hyperparameter tuning
- Cross-validation
- Larger medical imaging datasets

Such systems can support clinical decision making and improve diagnostic accuracy.